

ASCE Bridge Designer

Cloud Edition

...the newest version of a free, open source application available since 2002. Millions have used it to learn basics of engineering design. Along with other improvements, this version makes Bridge Designer more accessible. **Only a web browser equipped with a mouse-like pointer is needed. No software to install. Chromebooks work well.**

⇒ **Totally free. Secure and safe. No login. No ads. No PII collected.**

How does it work? Use your browser to start the app. The task is to design a highway bridge over a river gap. The app will guide you:

- **Choose from about 400 design conditions.** This determines the overall shape of your project.
- **Draw bridge trusses.** In a simplified CAD drafting board, choose joint locations, member geometry, and materials.
- **Push the test button.** The design appears in 3D. Walk around to inspect. It flexes as it absorbs weight. A truck drives over it. Colors show the flow of forces.
- **The truck either makes it across the gap, or the bridge fails.** An animation shows how failure occurs.
- **Iterate until your bridge supports the truck.** Make members stronger or position them better.
- **Now, minimize cost.** Iterate truss geometry and materials to support the truck *efficiently*.

Tradeoffs appear at every step. Exploring to find what works best is the heart of engineering!

How difficult is it? Any student able to use a web browser can design a working bridge in a few minutes. Finding an *optimal* bridge requires deep understanding of structural design. Learn using the comprehensive help system. Advanced features support deep dives for inspired users. Print a drawing or 3d scale model.

How can I teach it?

- Introduce the app in class and coach students through their first design.
- Conduct a contest. Let students explore their ideas with a fixed deadline. Lowest cost decides winners. Aesthetics can also be a component. The app includes contest support features.

More info? Write us at bridgedesignerteam@gmail.com.

Link? <https://asce.org/bridge-designer>

